

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular/Supplementary Winter Examination – 2024

Course: B. Tech

Branch: Common to all branches

Semester: III

Subject Code & Name:

Engineering Mathematics - III (BTBS301/BTES301/BTLOG301)

Max Marks: 60

Date: 05/02/2025

Duration: 3 Hr.

Instructions to the Students:

- Each question carries 12 marks.
- Question No. 1 will be compulsory and include objective-type questions.
- Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
- The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
- Use of non-programmable scientific calculators is allowed.
- Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	If $L\{f(t)\} = \frac{e^{-as}}{s^3}$ then $L\{f(3t)\}$ is equal to a. $\frac{e^{-s}}{(\frac{s}{a})^3}$ b. $\frac{e^{-s}}{(\frac{s}{3})^3}$ c. $\frac{27 e^{-\frac{as}{3}}}{s^3}$ d. None	Understand CO1	1
2	Laplace transform of the function $f(t) = e^{-3t} \cos 4t$ is, a. $\frac{s+3}{s^2+16}$ b. $\frac{s+3}{s^2+3}$ c. $\frac{s+3}{s^2+6s+25}$ d. None	Understand CO1	1
3	Laplace transform of the function $f(t) = t \sin at$ is, a. $\frac{2as}{(s^2-a^2)^2}$ b. $\frac{2s}{(s^2-a^2)^2}$ c. $\frac{2as}{s^2-a^2}$ d. None	Understand CO1	1
4	Inverse Laplace transform of the function $f(t) = \frac{15}{s^2+4s+13}$ is, a. $e^{-2t} \sin 3t$ b. $5 e^{-2t} \sin 3t$ c. $e^{-t} \sin 3t$ d. None	Understand CO2	1
5	Inverse Laplace transform of the function $f(t) = \frac{1}{\sqrt{s+4}}$ is a. $e^{-4t} \frac{1}{\sqrt{\pi t}}$ b. $e^{-t} \frac{1}{\sqrt{\pi t}}$ c. $e^{-4t} \frac{1}{\sqrt{t}}$ d. None	Understand CO2	1
6	The inverse Laplace transform of the function $f(t) = \frac{1}{s^2+9}$ is a. $\frac{1}{9} \sin 3t$ b. $\frac{1}{3} \sin 3t$ c. $\sin 3t$ d. None	Understand CO2	1
7	The Fourier cosine transform of e^{-x} is a. $\frac{s}{s^2+1}$ b. $\frac{1}{s^2+1}$ c. $\frac{s}{s^2-1}$ d. None	Understand CO3	1
8	The Fourier sine transform of e^{-ax} is a. $\frac{a}{a^2+s^2}$ b. $\frac{a}{a^2-s^2}$ c. $\frac{s}{a^2+s^2}$ d. None	Understand CO3	1
9	The partial differential equation obtained by eliminating a & b from $z = ax + by + ab$ is a. $z = xp + yq - pq$ b. $z = xp + yq + pq$ c. $z = xp - yq - pq$ d. None	Understand CO4	1
10	The Lagrange's linear partial differential equation is of the form a. $Pp - Qq = R$ b. $Pp + Qq = 0$ c. $Pp + Qq = R$ d. None	Understand CO4	1

11	If $f(Z) = u + iv$ in Polar form is analytic then $\frac{\partial u}{\partial r}$ is equal to				Understand CO5	1
	a. $\frac{\partial v}{\partial \theta}$	b. $r \frac{\partial v}{\partial \theta}$	c. $\frac{1}{r} \frac{\partial v}{\partial \theta}$	d. None		
12	If $f(z)$ is an analytic function with constant modulus, then $f(z)$ is a				Understand CO5	1
	a. constant function	b. harmonic function	c. Orthogonal	d. None		
Q. 2	Solve the following.					12
A)	Find the Laplace Transform of $F(t) = \frac{e^t - \cos t}{t}$				Apply/CO1	6
B)	Find the Laplace transform of $\int_0^t t e^{-t} \sin 4t dt$				Apply/CO1	6
Q.3	Solve the following.					12
A)	Using Partial Fraction method, find the inverse Laplace Transforms $\frac{5s+3}{(s-1)(s^2+2s+5)}$				Apply/CO2	6
B)	Solve $\frac{dy}{dt} + 2y = e^{-3t}$, $y(0) = 1$				Apply/CO2	6
Q. 4	Solve any TWO of the following.					12
A)	Find the Fourier transform of $f(x) = \begin{cases} 1, & \text{for } x < 1 \\ 0, & \text{for } x > 1 \end{cases}$. Hence evaluate that $\int_0^\infty \frac{\sin x}{x} dx$.				Apply/CO3	6
B)	Find the Fourier cosine transform of $f(x) = \frac{1}{1+x^2}$. Hence derive the Fourier sine transform of $\phi(x) = \frac{x}{1+x^2}$.				Apply/CO3	6
C)	Using Parseval's identity, show that $\int_0^\infty \frac{t^2}{(4+t^2)(9+t^2)} dt = \frac{\pi}{10}$.				Apply/CO3	6
Q.5	Solve any TWO of the following.					12
A)	Partial differential equation by eliminating the arbitrary function $z = x + y + f(xy)$				Understand CO4	6
B)	Solve $p(\tan x) + q(\tan y) = \tan z$				Apply /CO4	6
C)	Use the method of separation of variables to solve the equation $\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$.				Apply /CO4	6
Q. 6	Solve any TWO of the following.					12
A)	Find the analytic function whose imaginary part is $\frac{1}{2} \log(x^2 + y^2)$				Apply/CO5	6
B)	Show that function $v = \sinh x \cos y$ is harmonic function. Also find its harmonic conjugate function.				Remember CO5	6
C)	Apply Cauchy's integral Formula to evaluate $\oint_C \frac{\sin^2 z}{(z-\frac{\pi}{6})^2} dz$ where $C: z = 1$				Apply/CO5	6
	*****End*****					

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5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The Laplace transform of $F(t) = \operatorname{erf}(\sqrt{t})$ is equal to	Understand CO1	1
	a. $\frac{1}{s\sqrt{s^2+1}}$ b. $\frac{1}{s\sqrt{s-1}}$ c. $\frac{1}{s\sqrt{s+1}}$ d. None		
2	The Laplace transform of $F(t) = e^{2t} t^3$ is equal to	Understand CO1	1
	a. $\frac{12}{(s-4)^2}$ b. $\frac{12}{(s+4)^2}$ c. $\frac{12}{(s+2)^4}$ d. None		
3	The Laplace transform of $f(t) = t^{-\frac{1}{2}}$ is equal to	Understand CO1	1
	a. $\frac{2as}{(s^2-a^2)^2}$ b. $\frac{2s}{(s^2-a^2)^2}$ c. $\frac{2as}{s^2-a^2}$ d. None		
4	The inverse Laplace transform of $\bar{f}(s) = \frac{1}{(s+3)^5}$ is equal to	Understand CO2	1
	a. $\frac{e^{-3t}t^4}{24}$ b. $\frac{e^{3t}t^4}{24}$ c. $e^{-3t} t^4$ d. None		
5	The inverse Laplace transform of $\bar{f}(s) = \frac{s^2-3s+4}{s^3}$ is equal to	Understand CO2	1
	a. $1 - 3t - 2t^2$ b. $1 + 3t + 2t^2$ c. $1 - 3t + 2t^2$ d. None		
6	The inverse Laplace transform of the function $f(t) = \frac{1}{s^2+9}$ is	Understand CO2	1
	a. $\frac{1}{9}\sin 3t$ b. $\frac{1}{3}\sin 3t$ c. $\sin 3t$ d. None		
7	The Fourier cosine transform of the function $f(t)$ is	Understand CO3	1
	a. $F_c(s) = \int_0^\infty f(t) \cos st$ b. $F_c(s) = \int_0^\infty f(t) \cos t dt$ c. $F_c(s) = \int_0^\infty f(st) \cos t dt$ d. None		
8	The Fourier sine transform of e^{-ax} ($a > 0$) is	Understand CO3	1
	a. $\frac{s}{s^2+a^2}$ b. $\frac{a}{s^2+a^2}$ c. $\frac{s}{s^2-a^2}$ d. None		
9	The general solution of $zp = -x$ is given by	Understand CO4	1
	a. $\phi(x^2 + z^2, y) = 0$ b. $\phi(x^2 - z^2, y) = 0$ c. $\phi(x^2 + z^2, 2y) = 0$ d. None		
10	The Lagrange's linear partial differential equation is of the form	Understand CO4	1
	a. $Pp - Qq = R$ b. $Pp + Qq = 0$ c. $Pp + Qq = R$ d. None		

11	If $f(Z) = u + iv$ in Polar form is analytic then $\frac{\partial u}{\partial r}$ is equal to				Understand CO5	1
	a. $\frac{\partial v}{\partial \theta}$	b. $r \frac{\partial v}{\partial \theta}$	c. $\frac{1}{r} \frac{\partial v}{\partial \theta}$	d. None		
12	A Bilinear transformation map circles into				Understand CO5	1
	a. circles	b. hyperbola	c. parabola	d. None		
Q. 2	Solve the following.					12
A)	Find Laplace transform of $\frac{\cos 3t - \cos 2t}{t}$				Understand/ CO1	6
B)	Apply Laplace transform property to find Laplace transform of $\int_0^t te^{-4t} \sin 3t dt$				Apply/CO1	6
Q.3	Solve the following.					12
A)	Find inverse Laplace transform of $\log\left(\frac{s^2+b^2}{s^2+a^2}\right)$				Apply/CO2	6
B)	Solve $\frac{dy}{dt} + 2y = e^{-3t}$, $y(0) = 1$				Apply/CO2	6
Q. 4	Solve any TWO of the following.					12
A)	Find Fourier integral representation of the function $f(x) = \begin{cases} 1, & x < 1 \\ 0, & x > 1 \end{cases}$ and hence evaluate i) $\int_0^\infty \frac{\sin \omega \cos \omega x}{\omega} d\omega$ ii) $\int_0^\infty \frac{\sin \omega}{\omega} d\omega$				Apply/CO3	6
B)	Find the Fourier sine transform of $e^{- x }$, and hence show that $\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi e^{-m}}{2}$, $m > 0$.				Apply/CO3	6
C)	Using Parseval's identity, show that $\int_0^\infty \frac{t^2}{(4+t^2)(9+t^2)} dt = \frac{\pi}{10}$.				Apply/CO3	6
Q.5	Solve any TWO of the following.					12
A)	Obtain partial differential equation by eliminating arbitrary function f and g from $y = f(x - at) + g(x + at)$				Understand CO4	6
B)	Solve $yzp + zxq = xy$				Apply/CO4	6
C)	Use the method of separation of variables to solve the equation $\frac{\partial^2 u}{\partial x^2} - 2 \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$.				Apply /CO4	6
Q. 6	Solve any TWO of the following.					12
A)	Show that function $v = \sin x \cos y$ is harmonic function. Also find its harmonic conjugate function.				Remember /CO5	6
B)	Apply Cauchy's integral Formula to evaluate $\oint_C \frac{\sin^2 z}{(z-\frac{\pi}{6})^2} dz$ where $C: z = 1$				Apply/CO5	6
C)	State Cauchy's residue theorem and evaluate $\oint_C \frac{e^{-2z}}{(z-1)^2} dz$ where $C: z = 1.5$				Remember /CO5	6
	*****END*****					

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